

# CSCE 775: Deep Reinforcement Learning

## Homework #1

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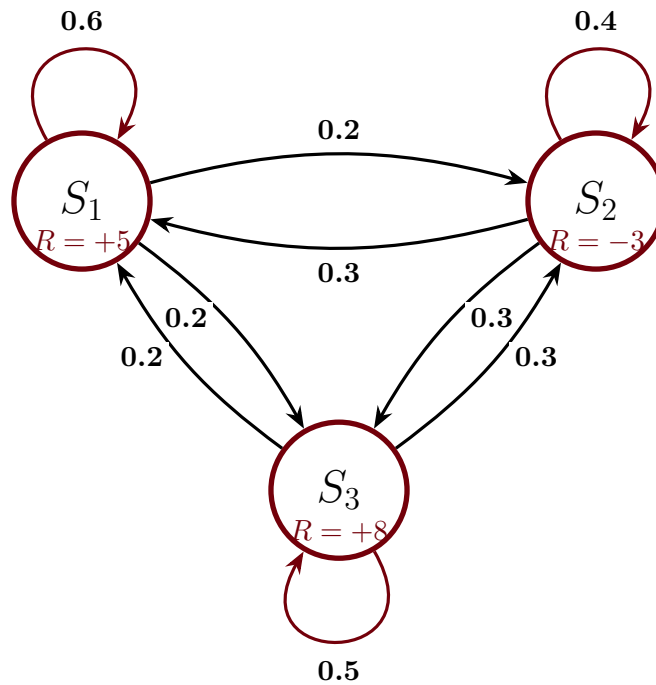
**Instructions:** Show all work. Write clearly and box final answers. You may use a calculator for arithmetic but all reasoning must be shown by hand.

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**Problem 1: MRP State Values**


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Consider the following Markov Reward Process with three states  $\mathcal{S} = \{S_1, S_2, S_3\}$ .



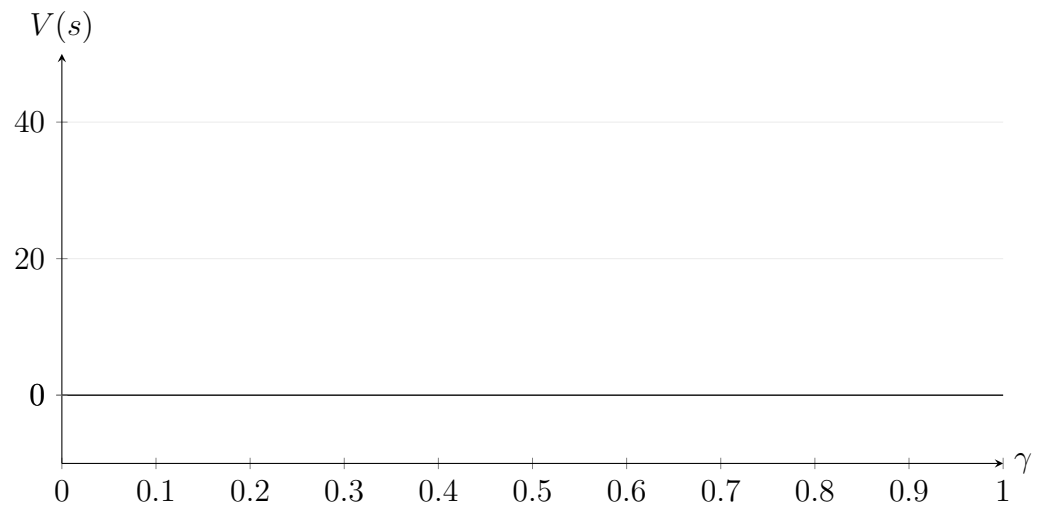
The transition matrix and reward vector are:

$$\mathcal{P} = \begin{pmatrix} 0.6 & 0.2 & 0.2 \\ 0.3 & 0.4 & 0.3 \\ 0.2 & 0.3 & 0.5 \end{pmatrix}, \quad \mathbf{R} = \begin{pmatrix} +5 \\ -3 \\ +8 \end{pmatrix}$$

- (a) Write the Bellman equation for this MRP in matrix form for a general discount factor  $\gamma$ . That is, express  $\mathbf{V}$  as a function of  $\mathbf{R}$ ,  $\gamma$ , and  $\mathcal{P}$ .
- (b) Form the matrix  $(\mathbf{I} - \gamma\mathcal{P})$  explicitly for  $\gamma = 0.8$ . Show all entries.

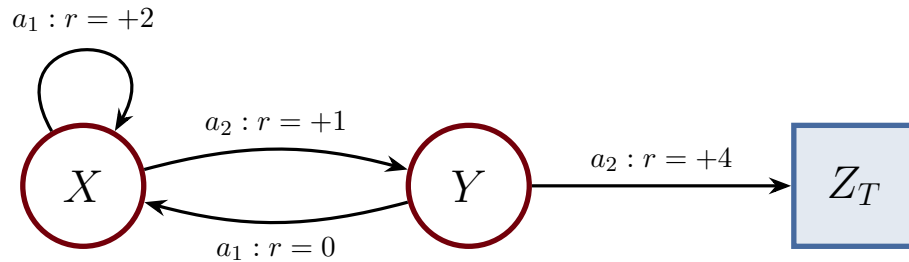
- (c) Solve the  $3 \times 3$  linear system  $(\mathbf{I} - \gamma\mathcal{P})\mathbf{V} = \mathbf{R}$  for  $\gamma = 0.8$  to find  $V(S_1)$ ,  $V(S_2)$ , and  $V(S_3)$ . Show your work (Gaussian elimination, Cramer's rule, or equivalent).

(d) On the axes below, sketch the curves  $V(S_1)$ ,  $V(S_2)$ , and  $V(S_3)$  as functions of  $\gamma \in [0, 1]$ . Label each curve and mark the zero-crossing from part (c).



### Problem 3: Value Iteration by Hand

Consider the following MDP with states  $\{X, Y\}$  and a terminal state  $Z_T$ , discount factor  $\gamma = 0.9$ . Each non-terminal state has two actions:  $a_1$  and  $a_2$ . All transitions are deterministic.



Transition and reward summary:

| State | Action | Next State | Reward |
|-------|--------|------------|--------|
| $X$   | $a_1$  | $X$        | $+2$   |
| $X$   | $a_2$  | $Y$        | $+1$   |
| $Y$   | $a_1$  | $X$        | $0$    |
| $Y$   | $a_2$  | $Z_T$      | $+4$   |

- (a) Initialize  $V_0(X) = V_0(Y) = 0$ . Perform **3 iterations** of value iteration. Fill in the table below. Show computation for each cell.

| Iter $k$ | $V_k(X)$ | $\pi_k(X)$ | $V_k(Y)$ | $\pi_k(Y)$ |
|----------|----------|------------|----------|------------|
| 0        | 0        | —          | 0        | —          |
| 1        |          |            |          |            |
| 2        |          |            |          |            |
| 3        |          |            |          |            |

- (b) After convergence, what is the optimal policy  $\pi^*$ ? State  $\pi^*(X)$  and  $\pi^*(Y)$  and justify briefly.
- (c) Compute the exact optimal values  $V^*(X)$  and  $V^*(Y)$  by solving the Bellman optimality equations simultaneously (not by iteration).

### Problem 4: Grid World Policy

Consider the  $3 \times 3$  grid MDP below. The agent can move **Up, Down, Left, Right**. Moves that would leave the grid keep the agent in place. All transitions are deterministic. Discount factor  $\gamma = 1$ .

|              |             |                          |
|--------------|-------------|--------------------------|
| (0, 2)       | (1, 2)      | <b>Goal</b><br>$r = +10$ |
| (0, 1)       | <b>Wall</b> | <b>Trap</b><br>$r = -5$  |
| <b>Start</b> | (1, 0)      | (2, 0)                   |

#### Rules:

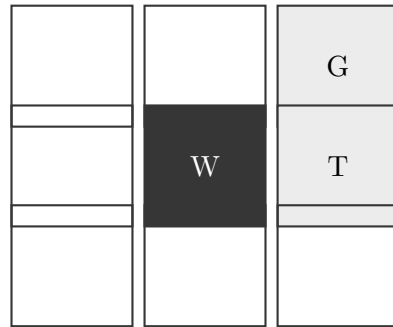
- Every move costs  $r = -1$  (step penalty), except moves into Goal or Trap.
- Reaching the **Goal** (2, 2): reward  $r = +10$ , episode ends.
- Reaching the **Trap** (2, 1): reward  $r = -5$ , episode ends.
- The **Wall** (1, 1) is impassable; moves into it keep the agent in place (still costs  $-1$ ).
- Goal and Trap are terminal states.

(a) List the reward  $R(s, a, s')$  for every non-terminal state  $s$  and every action  $a$ . Organize your answer clearly (a table is fine).

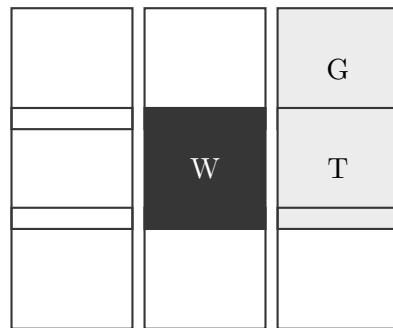
(b) Starting from  $V_0(s) = 0$  for all states, perform **2 iterations** of value iteration. Write the resulting values in each cell of the grids below.

**After Iteration 1:**

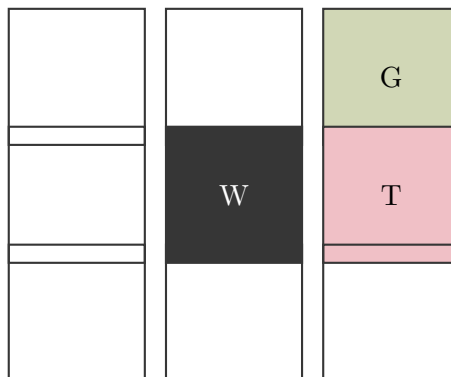




**After Iteration 2:**

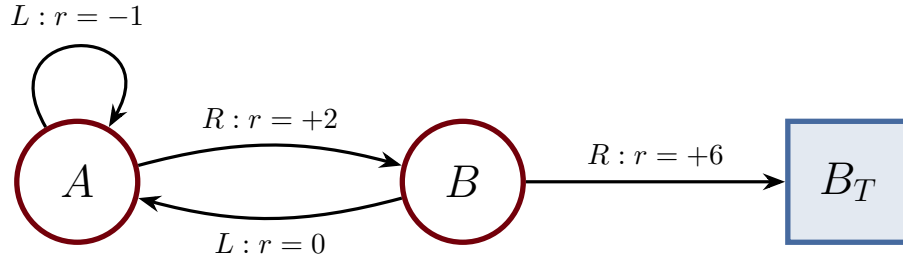


- (c) On the grid below, draw the optimal policy (one arrow per non-terminal cell indicating the best action).



### Problem 5: Q-Learning Episode Trace

Consider the following 2-state MDP with states  $\{A, B\}$  and terminal state  $B_T$ . Each state has two actions:  $L$  and  $R$ .



All transitions are deterministic. Use **Q-learning** with  $\alpha = 0.2$ ,  $\gamma = 1$ , and  $\varepsilon = 0$  (pure greedy; break ties alphabetically by action name, i.e.,  $L$  before  $R$ ).

Initialize all Q-values to zero:  $Q(A, L) = Q(A, R) = Q(B, L) = Q(B, R) = 0$ .

The three episode trajectories are given (not chosen by your policy—just apply the updates):

**Episode 1:**  $A \xrightarrow{R} B \xrightarrow{R} B_T$

**Episode 2:**  $A \xrightarrow{R} B \xrightarrow{R} B_T$

**Episode 3:**  $A \xrightarrow{L} A \xrightarrow{R} B \xrightarrow{R} B_T$

(a) For each step of each episode, compute the Q-learning update. Fill in the table.

Recall:  $Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$

| Ep. | Step | s | a | s'    | r  | New $Q(s, a)$ |
|-----|------|---|---|-------|----|---------------|
| 1   | 1    | A | R | B     | +2 |               |
| 1   | 2    | B | R | $B_T$ | +6 |               |
| 2   | 1    | A | R | B     | +2 |               |
| 2   | 2    | B | R | $B_T$ | +6 |               |
| 3   | 1    | A | L | A     | -1 |               |
| 3   | 2    | A | R | B     | +2 |               |
| 3   | 3    | B | R | $B_T$ | +6 |               |

(b) State all four Q-values after the 3 episodes are complete.

(c) What is the greedy policy  $\pi(s) = \arg \max_a Q(s, a)$  for each state?

**Problem 6: Exploration vs. Exploitation**

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- (a) Suppose an agent is in a state with three available actions  $\{a_1, a_2, a_3\}$  and current Q-values:

$$Q(s, a_1) = 5.0, \quad Q(s, a_2) = 3.2, \quad Q(s, a_3) = 7.1$$

Under an  $\varepsilon$ -greedy policy with  $\varepsilon = 0.4$ , compute the probability of selecting each action. Show your work.

- (b) Suppose  $\varepsilon$  starts at  $\varepsilon_0 = 1.0$  and decays multiplicatively each episode by a factor of 0.99:

$$\varepsilon_k = \varepsilon_0 \cdot 0.99^k$$

After how many episodes  $k$  is  $\varepsilon_k < 0.05$ ? Solve exactly using logarithms.

- (c) In 3–5 sentences, explain: Why is exploration important in the early stages of learning? Why is it beneficial to decay  $\varepsilon$  over time rather than keeping it fixed?